Data Science for Network Data: Approaches and Applications

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Artificial Intelligence & Data Science

Early Years 1940s ~1960s: Turing test

1974-1980: AI winter

1980s: Expert system, unable to learn from experience.

Statistics, Probability, Mathematics

Since 2000: the widespread adoption of the internet has led to an exponential increase in the generation of publicly accessible data. The terms '**big data**' and '**data science**' are relevant here, as they refer to **the analysis (or 'mining') of large datasets**, often by using AI to identify patterns and derive useful outputs from the data.

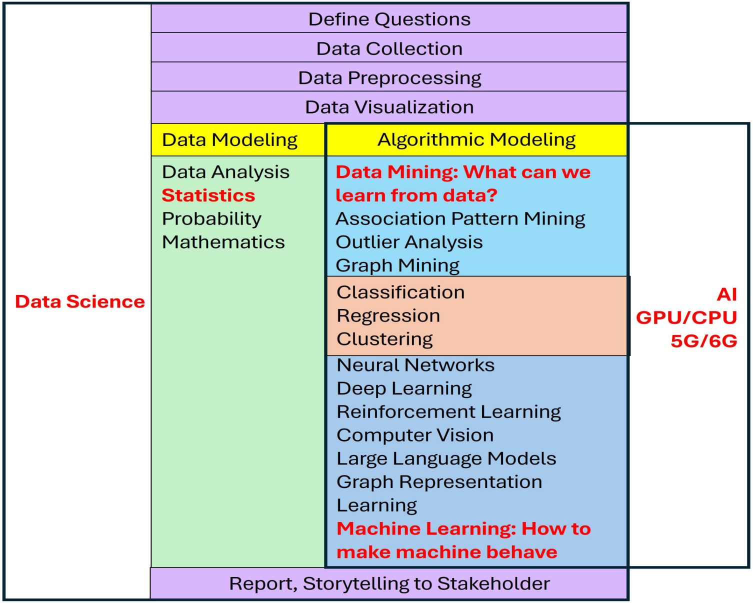
Big Data → **Innovation in AI**

Statistics, Probability, Mathematics

Data Mining

The Current State of AI: the term AI is becoming synonymous with Machine Learning, learn from examples, improve with experience.

Artificial Intelligence & Data Science

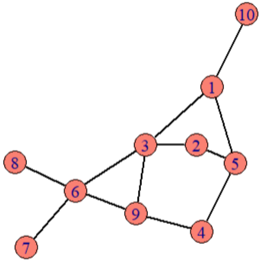


Network Graph

- A **Network or Graph** $\mathcal{G} = \{V, E\}$ is a set containing node set V and edge set E.
  The **Nodes** in V are the **entities**. The **Edges** in E represent the **interactions** between nodes.

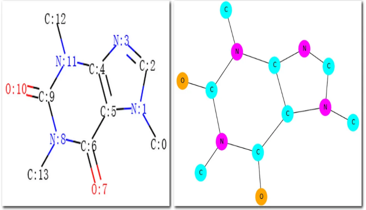
- The following is an example of graph $\mathcal{G} = \{V, E\}$, where

$$V = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \qquad E = \{13, 15, 110, 23, 25, 36, 39, 45, 49, 67, 68, 69\}$$

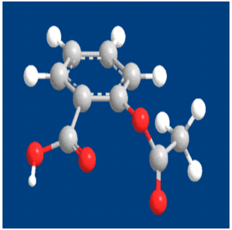


- real-world networks: Social Network, Molecule Graph, etc.

Network Graph

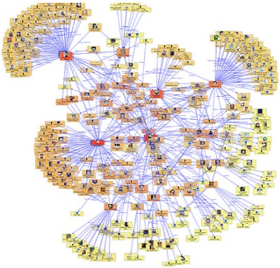


(1) Molecule Graph

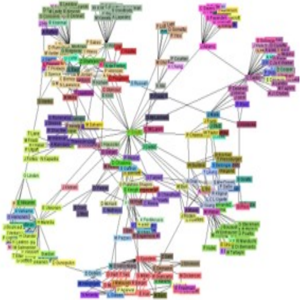


(2) Aspirin

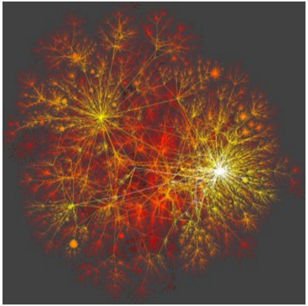
(3) Yeast protein interaction network



(4) Social Network



(5) Co-author Network



(6) Internet

Adjacency Matrix

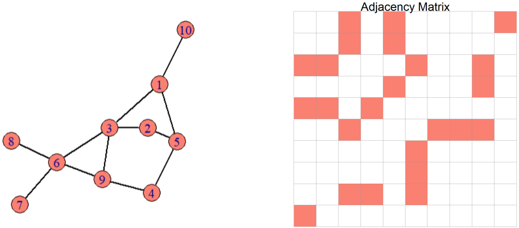
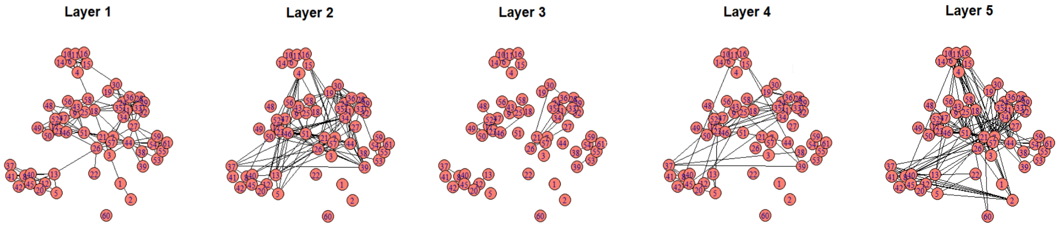


Figure: Graph $\mathcal{G} = \{V, E\}$ and its Adjacency Matrix $A_{n \times n}$

- Assumption on the graph: unweighted, undirected, no self-loops.
- Adjacency Matrix: $A_{ij} = 0$ or 1, symmetric, $A_{ii} = 0$.
- Edge Numbers $m = \frac{1}{2} \sum_{i,j} A_{ij}$
- Density: $D = \frac{m}{\binom{n}{2}} = \frac{2m}{n(n-1)}$ where m is edge numbers, n is node numbers.
  $[0, 0.01)$: very sparse; $[0.01, 0.1)$: sparse; $[0.1, 0.5)$: moderately dense; $[0.5, 1]$: dense.
- Node Degree $d_i = \sum_j A_{ij}$ Total Degree $d = \sum_{i,j} A_{ij} = 2m$ Average Degree $\bar{d} = \frac{d}{n} = \frac{2m}{n}$

Multilayer Networks: model different types of edges

- Multilayer Networks is a collection of networks that model complex systems by representing a fixed set of objects as nodes, and **captures various types of relationships** among them across different layers. Together these layers form the multilayer networks.
- Each layer encodes a specific mode of interaction or connection among the same set of nodes.
- **Why multilayer networks?**
  As real-world systems getting more complex, it is necessary to apply multilayer networks to capture diverse relationships among the objects in the system.



(1) having lunch together | (2) facebook connection | (3) co-authoring | (4) have leisure togerther | (5) work together

Figure: 5-Layer CS-Aarhus Networks $L = 5, n = 61$

Real-World Multilayer Networks: the CS-Aarhus networks

- Assumption of each layer/network, intra-layer edges:

unweighted, undirected, no self-loops

- Assumption of inter-layer edges:

each layer is independent with each other.

Entity / Node:	people				
Node size n:	61				
Layers L:	5				
Relationships of each layer:	A_1 : *have lunch together,*				
	A_2 : *facebook connection,*				
	A_3 : *co-authoring publication,*				
	A_4 : *have leisure together,*				
	A_5 : *work together.*				
Density of each layer:	A_1	A_2	A_3	A_4	A_5
	0.1055	0.0678	0.0115	0.0481	0.1060

Testing Common Subspace in Multilayer Networks

Common Subspace: definition

- Common properties or common subspace of multilayer networks is one active research problem.
- **How to interpret a common subspace in multilayer networks?**
  1)In the physical world, common subspace captures the **common structure** or **consensus community** across all layers;

  2)In mathematical language, common subspace is a **shared latent subspace** that can represent the **homogeneity** across all layers.
- With this **homogeneity** or **common structure**, we can **classify** or **cluster** the networks.

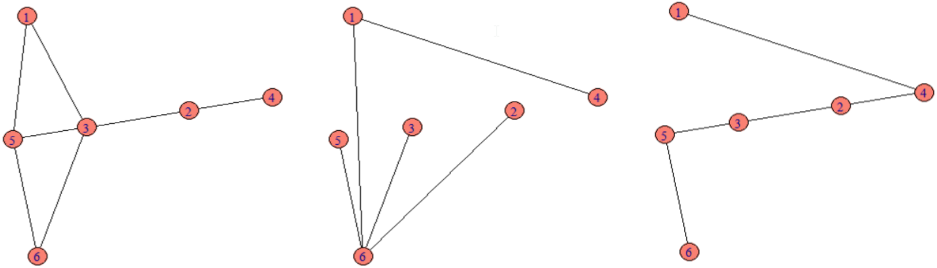


Figure: 3-Layer Multilayer Networks $n = 6$

Common Subspace: is there one existing?

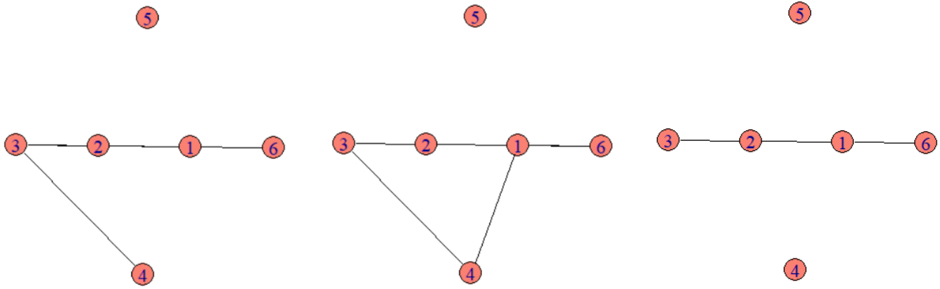


Figure: 3-Layer Multilayer Networks $n = 6$: share common subspace

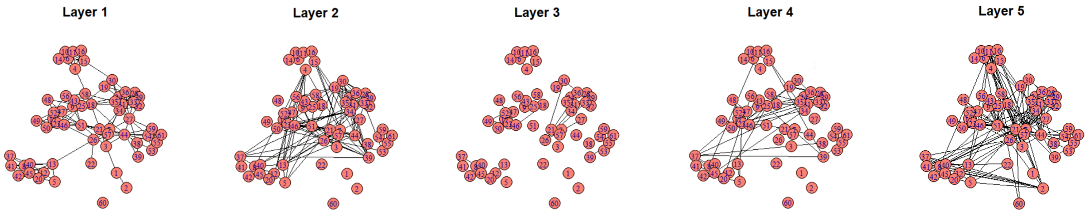


Figure: 5-Layer CS-Aarhus Networks $n = 61$: is there a **consensus community** across all layers?

Common Subspace: applications and literature review

- **In reality, we don't know** whether there is a shared common subspace in real-data of multilayer networks!

- Common subspace of the multilayer networks has been actively studied because it reveals shared community structures or homogeneity that have many **practical applications**.
  For example:
  1)multilayer social networks: consensus community;
  2)international trading networks: a robust clusters of trading partnerships;
  3)brain networks: reveal core connectivity patterns.

- Arroyo et al. 2021 proposed a spectral algorithm to estimate the common invariant subspace of multilayer networks.

- Pensky and Wang 2024, Pensky 2024 studied the problem of clustering multilayer networks into clusters/groups by their common subspace.

Testing Common Subspace in Multilayer Networks

- Given a collection of networks, Arroyo et al. 2021 assume that **All the networks** share common subspace; Pensky and Wang 2024 assume that **Some of the networks** share common subspace and some do not, and **the networks can be partitioned into clusters such that the networks within the same cluster have the common subspace**.

- A natural question is

Do ALL the networks or SOME of the networks share common subspace in multilayer networks?

- **I am the FIRST to formulate the problem as a hypothesis testing problem**

Under H_0: All networks sharing common subspace.

Under H_1: Some of the networks share the common subspace.

Random Graph Model

- A graph is said to be **random** if $A_{ij}(1 \leq i, j \leq n)$ are random.

Definition 1 ((Erdős–Rényi 1960; Bollobás et al. 2007))

Random heterogeneous graph model is defined as

$$A_{ij} \overset{\text{i.i.d.}}{\sim} Bernoulli(P_{ij}), \quad i < j.$$

$$P \text{ is } n \times n \text{ probability matrix. } P_{ij} \in [0, 1]. \quad P_{ii} = 0.$$

Assumption on Graph: unweighted, undirected, no self-loops,
it means adjacency matrix A: 1 or 0, $A_{ij} = A_{ji}$, $A_{ii} = 0$.

Model Assumption: $P_{ii} = 0$ satisfy $A_{ii} = 0$. A_{ij} are mutually independent.

$$\text{Example with 3 nodes: } P = \begin{bmatrix} 0.0, 0.8, 0.9 \\ 0.8, 0.0, 0.3 \\ 0.9, 0.3, 0.0 \end{bmatrix} \quad A = \begin{bmatrix} 0, 1, 1 \\ 1, 0, 0 \\ 1, 0, 0 \end{bmatrix}$$

- Expectation of adjacency matrix is $\mathbb{P}(A_{ij} = 1) = P_{ij}, \quad \mathbb{E}[A] = P$

Random Graph Model: Multilayer Networks

Definition 2 (Arroyo et al. 2021; Pensky and Wang 2024)

Given a positive integer L, multilayer networks $(A_1, A_2, \dots, A_L)$ follow the **Random Multilayer Heterogeneous Graphs Model** with each layer is mutually independent and can be factorized into

$$\mathbb{E}[A_l] = VQ_lV^T,$$

then two common subspaces is defined as

$$\mathbb{E}[A_l] = V^{(1)}Q^{(l)}(V^{(1)})^T, \quad 1 \le l \le L_1$$
$$\mathbb{E}[A_l] = V^{(2)}Q^{(l)}(V^{(2)})^T, \quad L_1 + 1 \le l \le L$$

where $A_{l,ij}$ are independent, $V^{(1)}, V^{(2)}$ are $n \times d$ column orthonormal matrices, $Q^{(l)}$ are square $d \times d$ matrix.

- Each A_l has different $Q^{(l)}$. Hence, they are heterogeneous.
- Orthonormal matrix V: $\text{span}\{v_1, v_2, \dots, v_d\}$ form the d-dimensional subspace.
- $A_1, A_2, \dots, A_{L_1}$ share the common subspace represented by $V^{(1)}$.
  $A_{L_1+1}, A_{L_1+2}, \dots, A_L$ share the common subspace represented by $V^{(2)}$.
  Multiple common subspaces $V^{(m)}, m \in [M], M \le L$ are similarly defined.
  $H_0 : V^{(1)} = V^{(2)} = \cdots = V^{(m)}, \quad H_1$: at least two are different.

Random Graph Model: the proposed model

- More generally, $\mathbb{E}[A_l] = V_l Q_l V_l^T$, $V_l \in \mathbb{R}^{n\times d}$ are obtained based on eigen-decomposition, its columns span a d-dimensional invariant subspace.
- Assume there is only one column in V_l spans one-dimensional invariant subspace.

Definition 3 (Yuan and Yao 2025)

Let W_l be a vector in $[0,1]^n$ such that $||W_l||_2 = 1$ for all $l \in [L]$, L is a positive integer, and ρ_l be a positive sequence that may depend on n. We say the multilayer networks $A_1, A_2, \ldots, A_L$ follow the **Rank-1 Random Multilayer Heterogeneous Graphs Model** $\mathcal{G}_n(W_1, W_2, \ldots, W_L)$ if

$$\mathbb{E}[A_l] = W_l \times \rho_l \times W_l^T \tag{1}$$

$$\mathbb{P}(A_{l,ij} = 1) = \rho_l W_{l;i} W_{l;j}, \quad i < j, \tag{2}$$

where $A_{l,ii} = 0$, $A_{l,ij} = A_{l,ji}$, $A_{l,ij}$ $(1 \le i < j \le n,\ 1 \le l \le L)$ are independent.

- $\mathbb{E}[d_i] = \sum_j \mathbb{E}[A_{ij}] = \rho ||W||_1 W_i \propto W_i$, W_l is a vector of the **degree-correction parameters** of A_l.
- rank(A) = number of linearly independent columns of A

  $\mathbb{E}[A_l] = [\rho_l w_1 W_l, \rho_l w_2 W_l, \ldots, \rho_l w_n W_l]$ contains one independent column. Thus, it is Rank-1.
- Each network is the **rank-1 degree-corrected Erdős-Rényi random heterogeneous graph** (Zhao et al. 2012; Gao et al. 2018).

Hypothesis Testing

- Given $A_1, A_2, \dots, A_L$ following the **Rank-1 Random Multilayer Heterogeneous Graphs Model** $\mathcal{G}_n(W_1, W_2, \dots, W_L)$, the following hypotheses are constructed.

$$H_0 : W_1 = W_2 = \cdots = W_L, \qquad H_1 : W_{l_1} \neq W_{l_2}, \text{ for some } l_1 \neq l_2. \tag{3}$$

  Under H_0, the graphs $A_1, A_2, \dots, A_L$ have the same degree-correction parameters.

  Under H_1, there exist at least two graphs such that their degree-correction parameters are different.

- **Weighted Degree Difference Test (WDDT test)** is proposed in my work for this hypothesis testing problem, which is the first to detect common subspace in multilayer networks.

Weighted Degree Difference Test (WDDT test)

$$D_n = \frac{1}{\sigma_n} \sum_{l=2}^{L} \left[\sum_{i=1}^{n} \left(\frac{d_{1,i}}{\sqrt{P_1}} - \frac{d_{l,i}}{\sqrt{P_l}} \right)^2 - \frac{d_1}{P_1} - \frac{d_l}{P_l} \right] \tag{4}$$

where $\sigma_n^2 = \frac{2(L-1)^2}{P_1} + \sum_{l=2}^{L} \frac{2}{P_l} + \sum_{l=2}^{L} \frac{4}{\sqrt{P_1 P_l}}$.

- node degree: $d_{l;i} = \sum_j A_{l,ij}$; total degree: $d_l = \sum_{i,j} A_{l,ij}$.
  $\mathbb{E}[A_{l,ij}] = \rho_l W_{l,i} W_{l,j}$, $\mathbb{E}[d_{l;i}] = \rho_l W_{l,i} ||W||_1$, $\mathbb{E}[d_l] = \rho_l \|W\|_1^2$.
- $P_l = \sum_{i \neq j \neq k} A_{l,ij} A_{l,jk}$.
- $\frac{d_{l,i}}{\sqrt{P_l}}$ are estimators of the degree-correction parameters $W_{l,i}$.
  The term $\sum_{i=1}^{n} \left(\frac{d_{1,i}}{\sqrt{P_1}} - \frac{d_{l,i}}{\sqrt{P_l}} \right)^2$ measures the difference between W_1 and W_l.
  $\frac{d_1}{P_1} + \frac{d_l}{P_l}$ centers it.
- σ_n^2 estimates the variance of D_n.

Weighted Degree Difference Test (WDDT test)

Assume: $\min_{1\le l\le L}\{\rho_l\} = \omega(1)$, $\|W_l\|_1 = \omega(1)$, $\rho_l = o(\|W_l\|_1)$, $\|W_l\|_4 = o(1)$; $\sum_i W_{1;i}W_{l;i} \le 1-\epsilon$.

Theorem 1 (Yuan and Yao 2025)

Under H_0, D_n converges in distribution to the standard normal distribution, as n goes to infinity. WDDT test is defined as:

$$\text{Reject } H_0 \text{ at significance level } \alpha \text{ when } |D_n| > u_{\frac{\alpha}{2}}.$$

- $u_{\frac{\alpha}{2}} = 1.96$ is the $100(1-\alpha/2)\%$ percentile of the standard normal distribution, if let $\alpha = 0.05$.
- Type I error of the WDDT test is asymptotically equal to α.

Theorem 2 (Yuan and Yao 2025)

Under H_1, the WDDT test statistics D_n has the following lower bound

$$D_n \ge \Theta\left(\frac{1}{r_n}\sum_{l=2}^{L}\left(1-\sum_{i=1}^{n} W_{1,i}W_{l,i}\right)\right)(1+o_P(1)),$$

where $r_n^2 = \frac{2(L-1)^2}{\rho_1^2\|W_1\|_1^2} + \sum_{l=2}^{L}\frac{2}{\rho_l^2\|W_l\|_1^2} + \sum_{l=2}^{L}\frac{4}{\rho_1\|W_1\|_1\rho_l\|W_l\|_1}$.

The order: $\frac{1}{r_n} \asymp (\rho_l\|W_l\|_1) \asymp D_n$. D_n goes to infinity in probability.

Simulation on Example 1

Example 1 (Yuan and Yao 2025)

- L layers multilayer networks:

$$\mathbb{P}(A_{l,ij} = 1) = \rho_l W_{l;i} W_{l;j}, \quad i < j$$

- Layers: $L \in \{2, 3, 4\}$; Node Size: $n \in \{200, 250, 300, 350, 400\}$
- $\rho_l = n^{\tau_l}$, $\tau_l \in \{0.1, 0.2, 0.3, 0.4\}$
- Degree-correction parameter $W_l = (W_{l,1}, W_{l,2}, W_{l,3}, ..., W_{l,n})$

$$W_{l,i} = \frac{\lambda_l \sqrt{r}}{\sqrt{n}} \quad \text{for } 1 \le i \le \frac{n}{r} \quad \text{and } W_{l,i} = \frac{\sqrt{\frac{r}{r-1}(1-\lambda_l^2)}}{\sqrt{n}} \quad \text{for } \frac{n}{r} < i \le n.$$

  $r =\in \{1.5, 2, 2.5, 3\}$ satisfying $r > 1$, $\lambda_l \in \{0.8, 0.7, 0.6, 0.5\}$ satisfying $0 < \lambda_l \le 1$.

- Indication for a layer: $A_l \sim \mathcal{G}_n(\tau_l, \lambda_l)$
- $Difference = \sum_l |\lambda_1 - \lambda_l|, \quad$ where $\lambda_l \in \boldsymbol{\lambda}$

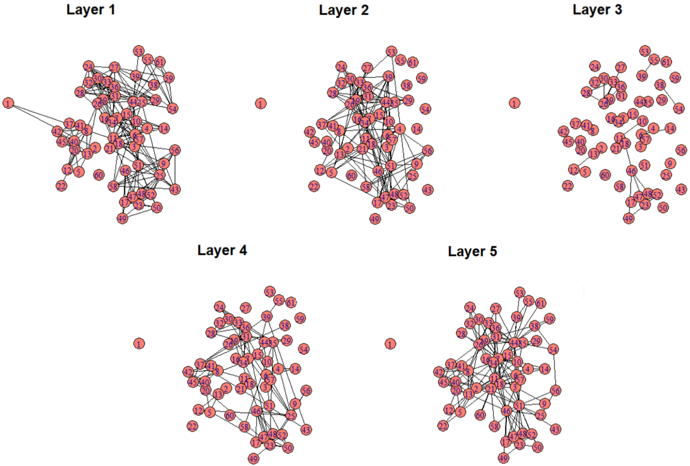
Simulation on Example 1: WDDT test

WDDT test: Three-Layer Networks $L = 3$				
$\boldsymbol{\tau}$	(0.3, 0.2, 0.4)	(0.3, 0.2, 0.4)	(0.3, 0.2, 0.4)	(0.3, 0.2, 0.4)
$\boldsymbol{\lambda}$	(0.8, 0.8, 0.8)	(0.8, 0.7, 0.6)	(0.8, 0.7, 0.5)	(0.8, 0.6, 0.5)
Hypotheses	$W_1 = W_2 = W_3$	$W_1 \neq W_2 \neq W_3$	$W_1 \neq W_2 \neq W_3$	$W_1 \neq W_2 \neq W_3$
$Difference$	0	0.3	0.4	0.5
$\boldsymbol{r = 1.5}$				
$n = 200$	0.054	0.199	0.471	0.763
$n = 250$	0.045	0.267	0.644	0.878
$n = 300$	0.055	0.361	0.790	0.956
$n = 350$	0.046	0.465	0.890	0.987
$n = 400$	0.051	0.541	0.945	0.999
$\boldsymbol{r = 2}$				
$n = 200$	0.052	0.186	0.567	0.785
$n = 250$	0.053	0.246	0.721	0.899
$n = 300$	0.054	0.368	0.862	0.972
$n = 350$	0.048	0.486	0.953	0.999
$n = 400$	0.052	0.568	0.964	1.000
$\boldsymbol{r = 2.5}$				
$n = 200$	0.051	0.167	0.617	0.796
$n = 250$	0.050	0.266	0.774	0.925
$n = 300$	0.048	0.353	0.907	0.969
$n = 350$	0.045	0.457	0.955	0.996
$n = 400$	0.053	0.584	0.982	1.000
$\boldsymbol{r = 3}$				
$n = 200$	0.059	0.167	0.593	0.811
$n = 250$	0.044	0.252	0.792	0.935
$n = 300$	0.057	0.378	0.893	0.985
$n = 350$	0.056	0.460	0.968	0.997
$n = 400$	0.054	0.549	0.991	1.000

- The simulated **type I errors** are close to 0.05.
- The simulated powers approach to 1.
- WDDT test has satisfactory performance.

Real Data Application: the CS-Aarhus networks

Layers	Association	Density
A_1	Representative of two individuals having Lunch together	0.105
A_2	Representative of two individuals having a social connection via Facebook	0.068
A_3	Representative of two individuals co-authoring a publication	0.011
A_4	Representative of two individuals having Leisure together	0.048
A_5	Representative of two individuals working together	0.106



Simulation on Real Data: WDDT test

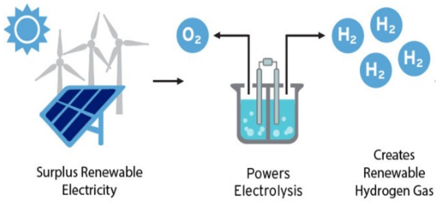
Layers	WDDT test	P-Value	Conclusion
A_1, A_2, A_3, A_4, A_5	5.921	0.000	$Reject H_0$
A_2, A_3, A_4, A_5, A_1	7.436	0.000	$Reject H_0$
A_3, A_4, A_5, A_1, A_2	3.081	0.002	$Reject H_0$
A_4, A_5, A_1, A_2, A_3	6.764	0.000	$Reject H_0$
A_5, A_1, A_2, A_3, A_4	8.323	0.000	$Reject H_0$
A_1, A_2, A_3, A_4	4.973	0.000	$Reject H_0$
A_1, A_2, A_3, A_5	5.207	0.000	$Reject H_0$
A_1, A_2, A_4, A_5	6.410	0.000	$Reject H_0$
A_1, A_3, A_4, A_5	4.304	0.000	$Reject H_0$
A_2, A_3, A_4, A_5	5.889	0.000	$Reject H_0$
A_1, A_4, A_5	3.634	0.000	$Reject H_0$
A_1, A_3, A_5	3.512	0.000	$Reject H_0$
A_1, A_3, A_4	3.316	0.001	$Reject H_0$
A_1, A_2, A_5	5.488	0.000	$Reject H_0$
A_1, A_2, A_4	4.823	0.000	$Reject H_0$
A_1, A_2, A_3	4.214	0.000	$Reject H_0$
A_2, A_4, A_5	6.462	0.000	$Reject H_0$
A_2, A_3, A_5	3.974	0.000	$Reject H_0$
A_2, A_3, A_4	4.350	0.000	$Reject H_0$
A_3, A_4, A_5	1.616	0.106	***Not Reject H_0***
A_1, A_2	3.641	0.000	$Reject H_0$
A_1, A_3	2.472	0.013	$Reject H_0$
A_1, A_4	1.811	0.070	***Not Reject H_0***
A_1, A_5	2.234	0.025	$Reject H_0$
A_2, A_3	2.344	0.019	$Reject H_0$
A_2, A_4	3.816	0.000	$Reject H_0$
A_2, A_5	3.233	0.001	$Reject H_0$
A_3, A_4	0.818	0.413	***Not Reject H_0***
A_3, A_5	0.819	0.413	***Not Reject H_0***
A_4, A_5	3.723	0.000	$Reject H_0$

Molecular Representation for Catalyst Discovery by Machine Learning

Renewal Energy Usage by Hydrogen

(1) Polar ice caps are melting as global warming

(2) Rising Greenhouse Gas Emission



(3) Hydrogen Energy Storage

- Challenges of climate change and energy scarcity
- Conventional energy resources are finite
- Renewable energy is one way out. There are more and more usage of renewable energy, such as wind, solar, hydrogen.
- Electricity can be generated in the reaction of Hydrogen gas:

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l) \ + \ 572 \text{ kJ } (286 \text{ kJ/mol})$$

- In order to make the process efficient for large-scale applications, effective catalysts are needed.

Molecules and Density Functional Theory (DFT)

- Catalysts are molecules.
- **Chemoinformatics** is the field that uses computational methods to represent, store, and analyze chemical data.
- **Schrodinger Equation** describes the system of atoms or nuclei and electrons.

$$\hat{H}\,\Psi = E\,\Psi,$$

  $\hat{H}$ = Hamiltonian operator (total energy operator)
  Ψ = wave function (state of the system)
  E is the measurable energy of the system when it is in state Ψ

- DFT replace wave function with electron density.
- **Hohenberg-Kohn Theorem** are the foundation of DFT.
  1) The ground-state electron density uniquely determines the external potential (and thus the Hamiltonian and all properties),
  2) The true ground-state density is the one that minimizes the energy functional.
- For non-quantum chemistry audiences, the key takeaway is that DFT calculate the lowest possible electronic energy configuration of that molecule at fixed geometry.

Molecules and Density Functional Theory (DFT)

- QM7 contains molecules with energies computed by DFT.
- Physical intuition: The molecule sits in a very deep energy well (large negative number).
  Total energy (larger absolute value) = baseline "depth of the well";
  Atomization energy (smaller absolute value) = "height needed to escape the well";

```
Name
0001.xyz
0002.xyz
0003.xyz
0004.xyz
0005.xyz
0006.xyz
0007.xyz
0008.xyz
0009.xyz
0010.xyz
```

(1) Data set of list of molecules by DFT

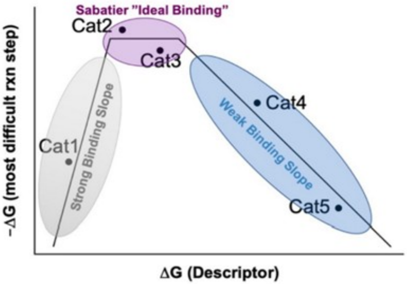
```
0001.xyz
5
charge = 0
 C 1.041682 -0.056200 -0.071481
 H 2.130894 -0.056202 -0.071496
 H 0.678598 0.174941 -1.072044
 H 0.678613 0.694746 0.628980
 H 0.678614 -1.038285 0.228641
```

(2) Properties of each molecule calculated by DFT

```
qm7/0001.xyz -417.031 -431.787271961
qm7/0002.xyz -711.117 -730.835894942
qm7/0003.xyz -563.084 -573.10424863
qm7/0004.xyz -403.695 -412.90228872
qm7/0005.xyz -858.499 -869.887691974
qm7/0006.xyz -1006.61 -1029.76951291
qm7/0007.xyz -860.212 -874.908818787
qm7/0008.xyz -707.023 -719.488803099
qm7/0009.xyz -724.049 -795.464941849
qm7/0010.xyz -876.545 -964.085902663
```

(3) Energy of each molecule

Predict DFT Energy to Discover Catalyst Candidate

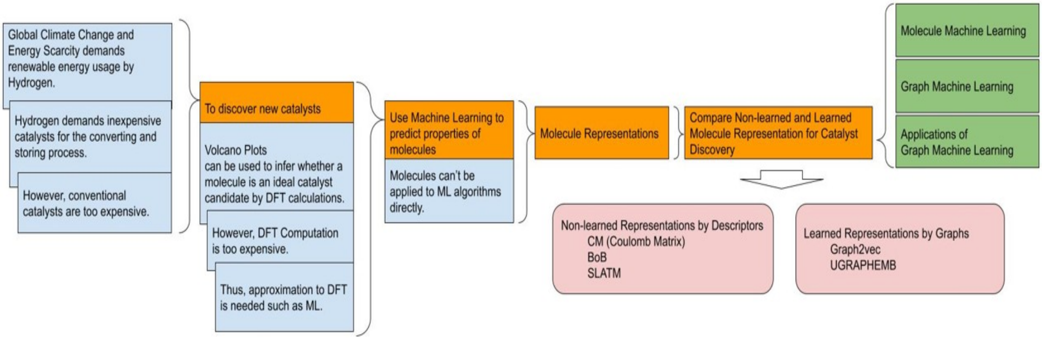


(1) Data set of list of molecules by DFT

- Use volcano plot to guide catalyst screening by DFT-calculated descriptors.
- Volcano plot consists three regions, x-axis is the descriptor of DFT computed energy.
- Sabatier's principle: an ideal catalyst should bind intermediates neither too strong nor too weak.
- The volcano plateau (or the peak) at Cat2 and Cat3 indicates catalyst bind intermediates neither too strong nor too weak, therefore, can be regarded as ideal catalysts candidates.
- In summary, once we can compute DFT-based descriptors, we can directly use the volcano plot to guide catalyst screening.

Motivation

- DFT is computationally expensive.
- Transform DFT computation task to machine learning prediction task.
- **Motivation and Pipeline of Computation**:



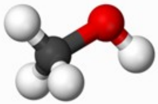
- Goal: to **represent molecules** as inputs for predictive models.
- QM7 is useful to help Machine Learning Models learn chemical representations

Molecular Representations: Non-learned vs Learned Approaches

Non-learned: Molecular Descriptors
The problem of how to **represent a molecule or material** has been a topic dating back to many decades ago and the wealth of information (and opinions) about this subject is well manifested by the collection of descriptors compiled in Todeschini and Consonni's Handbook of molecular descriptors (Todeschini and Consonni 2008).

Molecular Descriptor is the final result of a logic and mathematical procedure which transforms chemical information encoded within a symbolic representation of a molecule into a useful number.

- Coulomb Matrix
- Bag of Bonds
- Spectrum of Landon Axilrod-Teller-Muto potential (SLATM)



$$\begin{bmatrix} 36.9 & 33.7 & 5.5 & 3.1 & 5.5 & 5.5 \\ 33.7 & 73.5 & 4.0 & 8.2 & 3.8 & 3.8 \\ 5.5 & 4.0 & 0.5 & 0.35 & 0.56 & 0.56 \\ 3.1 & 8.2 & 0.35 & 0.5 & 0.43 & 0.43 \\ 5.5 & 3.8 & 0.56 & 0.43 & 0.5 & 0.56 \\ 5.5 & 3.8 & 0.56 & 0.43 & 0.56 & 0.5 \end{bmatrix}$$

Figure: Example of Coulomb Matrix for Methanol: CH3OH

Molecular Representations: Non-learned vs Learned Approaches

Learned: Graph Machine Learning

Non-learned featurizations demands domain experts to manually choose features, which cause some questions for researchers: What features should be generated for the data?

- UGRAPGEMB: Unsupervised Graph-level Embedding (Bai, Y., et al. 2019)
- Graph2vec (Narayanan, A., et al. 2017)

Wordembedding Featurized Representation

a	apple	...	year	unknow
f1	.	.	.	.
.	.	.	.	.
f300	.	.	.	.

300X10,000 x One-Hot Vector O_j 10,000X1 = w_j 300X1

Word2Vec

Data Science is Great.

Pairs:
[target word, context word]
[data, science]
[science, data]
[science, is]
[is, science]
[is, great]
[great, is]

Initialize Vectors a sequence of words:
V_data;
V_science;
V_is;
V_great;

$\Pr(\omega_c|\omega_t)$ the probability of visiting word c starting from word t

$$\Pr(\omega_c|\omega_t) = \frac{\exp(u_t^T \cdot v_c)}{\sum_{j=0}^{N}(\exp(u_t^T \cdot v_j))}$$

Learning Objective: Maximize($\prod_{all\ pairs} \Pr(\omega_c|\omega_t)$)

Graph2Vec

G1, G2, G3

Pairs
[graph, subgraphs]
[G1, subgraph1_1]
[G1, subgraph1_2]
[G1, subgraph1_3]
[G2, subgraph2_1]
[G2, subgraph2_2]
[G3, subgraph3_1]
[G3, subgraph3_2]

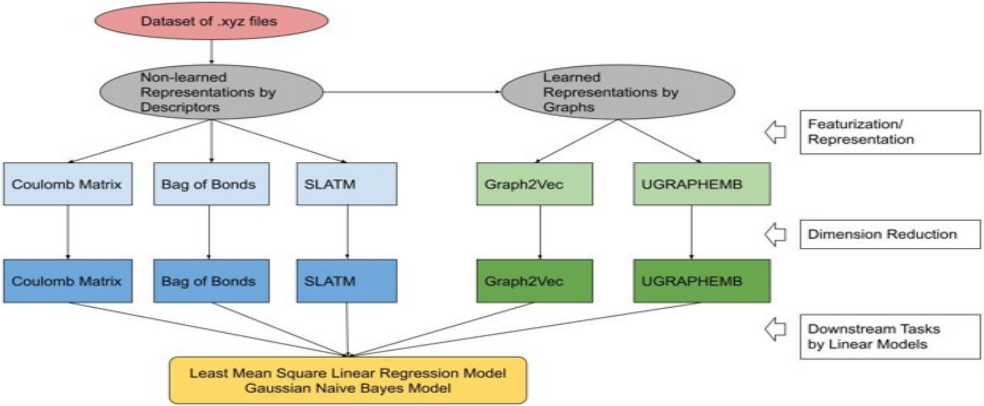
Initialize Vectors for a sequence of subgraphs:
Z_G1;
Z_G2;
Z_G3;
U_sg1_1;
U_sg1_2;
U_sg1_3;
U_sg2_1;
U_sg2_2;
U_sg3_1;
U_sg3_2;

$\Pr(r_{sg}|g)$ the probability of subgraph given the graph

$$\Pr(r_{sg}|g) = \frac{\exp(Z_G^T \cdot U_{sg})}{\sum_{j=0}^{N}(\exp(Z_G^T \cdot U_j))}$$

Learning Objective : Maximize($\sum all\ pairs \log(\Pr(r_{sg}|g))$)

Computational Pipeline



CM	BoB	SLATM	Graph2Vec	UGRAPGEMB
$[m, 276]$	$[m, 465]$	$[m, 10552]$	$[m, 128]$	$[m, 96]$

Table: m Molecules Embedded into Matrix

- Computing Tools: *qml, xyz2graph, Graph2Vec in Karateclub, StellarGraph, Tensorflow, Keras, Networkx, ScikiLearn, Numpy, Pandas, Matplotlib, Python*

Results

RMSE	Non-learned Representations by Descriptors			Learned Representations by Graphs	
Linear Regression	CM	BoB	SLATM	Graph2vec	UGRAPHEMB
D = 10	63	137	68	260	252
D = 30	50	25722	48	263	230
D = 50	51	26497	183	265	230
D = 70	48	31583	185	282	225
D = 90	69	38219	33	275	230

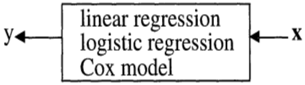
Figure: RMSE Results for Linear Regression

- Coulomb Matrix (CM) performs best among all the five types of representations (CM, BoB, SLATM, Graph2vec, UGRAPHEMB).
- Graph2vec and UGRAPHEMB, which are learned representations, perform stably regardless which linear models is used and regardless the dimension of the representations. Thus, we can say that the learned representations are more stable than the non-learned representations.

Two Cultures: Statistical Methods and Machine Learning

Statistical Methods: Data Modeling Culture

- Data Models
- All the articles in the flagship journal of theoretical statistics started with: Assume that the data are generated by the following model... Followed by mathematics exploring inference, hypothesis testing and asymptotics.

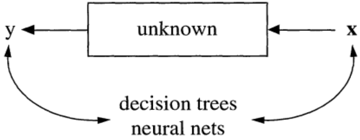


response variable = f(predictor variables, random noise, parameters)

- The conclusions are about the model's mechanism, and not about nature's mechanism.
- However, is it easy to invent a reasonable good parametric class of models for a complex mechanism devised by nature?
- Data modeling produces a simple and understandable picture of the relationship between the input variables and responses.

Machine Learning: Algorithmic Modeling Culture

- Predictive Accuracy for Real Data
- Accept that nature produces data in a black box whose insides are complex, mysterious, and, at least, partly unknowable.



- What is observed is a set of $x's$ that go in and a subsequent set of $y's$ that come out. The problem is to find an algorithm $f(x)$ such that for future x in a test set, $f(x)$ will be a good predictor of y.
- Shifts focus from data models to properties of algorithms
- Accuracy vs. Interpretability?

Future Work

- About Interpretbility: AI + Statistics
- To uncover the underlying mechanisms of how systems work and interpret the results.

- About Innovation: where zero-to-one occurs?
- Beyond technical proficiency, Where does true innovation, the transition from zero to one, actually occur?
- Solve the problem, make real world impact.

- Testing Common Subspace in Multilayer Networks
- Consider more complex models with higher Rank-q to model multilayer networks;
- Empirical Likelihood test performs satisfactory with corresponding higher powers than WDDT test based on simulation results. A theoretical proof of its asymptotic distribution is needed.

- Molecular Representations for Property Prediction by Machine Learning
- Apply more advanced learning models and explore more use cases
- Train the model based on more available dataset to improve accuracy.

Q & A

END

Thank You

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Advance Techonoliges to Solve Society's Major Problems

Development of Technologies

- Artificial Intelligence
  Computer Vision, Natural Language ProcessingNLP), Reinforcement Learning, Edge AI / Explainable AI, Robotics, Autonomous Driving, Virtual Assistant , LLM(ChatGPT, Gemini), Machine Learning
- Connectivity & Computing
  5G/6G / Next-Gen Wireless / Neuromorphic Computing / Cloud & Edge Computing / High Performance Computing (HPC)
- Data Science / Machine Learning
  Machine Learning, Neural Networks, Predictive Models, Statistics, Data Processing, Data Analytics, Data Visualization
- Emerging & Cross-cutting Tech
  Metaverse / Virtual Reality / Augmented Reality, Digital Twins, Synthetic Biology & Bioinformatics, Cybersecurity AI, Human-Computer Interaction (HCI) / Brain-Computer Interface

Major Topics Globally

- Energy and Invironment
  Energy scarcity / Climate change / Global warming / Pollution & Air quality / Water scarcity / Renewable energy integration / Waste management & Recycling
- Economy and Society
  Global trading & supply chain risks, Income inequality / Poverty, Inflation & Financial instability, Unemployment & Job displacement, Cybercrime / Fraud detection,
- Health and Healthcare
  Healthcare accessibility & affordability, Aging population & chronic diseases, Pandemics & infectious diseases, Mental health challenges, Personalized medicine / Targeted interventions,
- Data, AI, and Systems
  Anomaly detection & system optimization, Behavioral patterns analysis, Privacy & data security, Algorithmic bias & fairness, AI model robustness & interpretability, Cyber-physical system vulnerabilities
- Societal and Technological Challenges
  Urban congestion / Smart city management, Transportation & autonomous mobility, Misinformation / Fake news, Education accessibility & quality, Ethical dilemmas in AI & biotech

Artificial Intelligence & Data Science

Artificial Intelligence:

- **Early Years 1940s ~1960s**: Turing test
- **1974-1980**: AI winter
- **1980s**: Expert system, unable to learn from experience.
- **Since 2000s**, the widespread adoption of the internet has led to an exponential increase in the generation of publicly accessible data. The terms '**big data**' and '**data science**' are relevant here, as they refer to the **analysis (or 'mining') of large datasets**, often by using AI to identify patterns and derive useful outputs from the data.
- **The Current State of AI**: Nowadays, the term AI is becoming synonymous with Machine Learning, learn from examples, improve with experience.
- In university teaching activities, Data Mining and Machine Learning are taught as the same course, and Big Data and Data Science are considered equivalent.

Data Science:

1. **Define the problem within domain**
   - Problem definition
   - Data collection
   - Data cleaning and preprocessing
   - Data Visualization: Tableau / Power BI / Python/ R
   - Exploratory Data Analysis (EDA)

   Business sense and domain knowledge are important
2. **Data Modeling & Algorithmic Modeling**
   - Dimension Reduction
   - Statistical Inference
   - Machine Learning (Data Mining)
   - Model training/ Model evaluation/ Deployment
   - Monitoring and maintenance

   This part includes data mining, machine learning, statistical inference, programming languages and computing software
3. **Share insights with stakeholders**
   - Reporting, Dashboard creation

Lower bound of D_n

Basic Math Notation

Order of the convergence and Convergence rate

$a_n = n^\alpha, \alpha > 0$, then $a_n \to \infty$ as $n \to \infty$ $a_n = \Theta(b_n)$ if $c_1 b_n \leq a_n \leq c_2 b_n$

$a_n = \omega(b_n)$ if $lim_{n\to\infty} \frac{a_n}{b_n} = \infty$

$a_n = O(b_n)$ if $a_n \leq c b_n$.

$a_n = o(b_n)$ if $lim_{n\to\infty} \frac{a_n}{b_n} = 0$

$X_n = O_p(a_n)$ if $\frac{X_n}{a_n}$ is bounded in probability.

$X_n = o_p(a_n)$ if $\frac{X_n}{a_n}$ converges to 0 in probability.

Assumptions

L-p norm

$||W||_p = (\sum_1^n w_i^p)^{\frac{1}{p}}$; $||W||_1 = \omega(1)$, $||W||_2 = 1$; $||W||_3 = o(1)$, $||W||_4 = o(1)$,

$min\{\rho_l\} = \omega(1)$, meaning $\rho_l \to \infty$

$\rho_l = o(||W||_1)$, converge to ∞ much slower than $||W||_1$.

result = mean(abs(Dn) > 1.96)

$\alpha = P(|D_n| \geq 1.96)$ under H_0

$\beta = P(|D_n| \geq 1.96)$ under H_1

Lower bound of D_n

$$D_n = \frac{1}{\sigma_n}\sum_{l=2}^{L}\left[\sum_{i=1}^{n}\left(\frac{d_{1,i}}{\sqrt{P_1}} - \frac{d_{l,i}}{\sqrt{P_l}}\right)^2 - \frac{d_1}{P_1} - \frac{d_l}{P_l}\right],$$

$$\text{where } \sigma_n^2 = \frac{2(L-1)^2}{P_1} + \sum_{l=2}^{L}\frac{2}{P_l} + \sum_{l=2}^{L}\frac{4}{\sqrt{P_1 P_l}}, \quad P_l = \sum_{i\neq j\neq k} A_{l,ij}A_{l,jk}$$

- $d_{l;i} = \sum_j A_{l,ij} \quad d_l = \sum_{i,j} A_{l,ij}$.
- $||W||_q = (\sum_1^n w_i^q)^{\frac{1}{q}}; ||W||_1 = \omega(1), ||W||_2 = 1; ||W||_3 = o(1), ||W||_4 = o(1),$
  $\rho_l = o(||W||_1), min\{\rho_l\} = \omega(1)$, meaning $\rho_l \to \infty$, $lim \frac{\rho_l}{||W||_l} \to 0$
- $E[A_{l,ij}] = \rho_l w_{l,i} w_{l,j}$, $\bar{A_{l,ij}} = A_{l,ij} - E[A_{l,ij}]$
- Example of how $||W||$ L-Norm used.
  $\sum \rho_l w_{l,i} w_{l,j}^2 w_{l,k} = \rho_l ||W||_1 ||W||_2 ||W||_1$; $\sum E[A_{l,ij}]E[A_l, jk] = \sum \rho_l w_{l,i} w_{l,j} \rho_l w_{l,j} w_{l,k} = \rho_l^2 ||W||_1^2$
- order of $P_l = \rho_l^2 ||W_l||^2$
  order of $\sigma_n^2 = \frac{1}{\rho_l^2 ||W_l||^2}$

  leading term in T_n
  $\sum_{i=1}^{n}\left(\frac{d_{1,i}}{\sqrt{P_1}} - \frac{d_{l,i}}{\sqrt{P_l}}\right)^2 - \frac{d_1}{P_1} - \frac{d_l}{P_l} = \Theta(\sum_{i=1}^{n}\left(\frac{E[d_{1,i}]}{\sqrt{P_1}} - \frac{E[d_{l,i}]}{\sqrt{P_l}}\right)^2 = \Theta((1 - \sum_{i=1}^{n} W_{1,i}W_{l,i})) \geq \Theta(1);$
  all other terms are $o(1)$
  Thus, $D_n = \frac{T_n}{\sigma_n} \geq \Theta\left(\frac{1}{\sigma_n}\sum_{l=2}^{L}\left(1 - \sum_{i=1}^{n} W_{1,i}W_{l,i}\right)\right)(1 + o_P(1))$

Asymptotic Distribution of D_n

From D_n to Z_n

$$D_n = \frac{1}{\sigma_n}\sum_{l=2}^{L}\left[\sum_{i=1}^{n}\left(\frac{d_{1,i}}{\sqrt{P_1}} - \frac{d_{l,i}}{\sqrt{P_l}}\right)^2 - \frac{d_1}{P_1} - \frac{d_l}{P_l}\right],$$

$$D_n = Z_n + o_P(1)$$

$$Z_n = \frac{1}{\sigma_n}\sum_{l=2}^{L}\left(\frac{1}{P_1}\sum_{i\neq j\neq k}\bar{A}_{1,ij}\bar{A}_{1,ik} + \frac{1}{P_l}\sum_{i\neq j\neq k}\bar{A}_{l;ij}\bar{A}_{l,ik} - \frac{2}{\sqrt{P_1P_l}}\sum_{i\neq j\neq k}\bar{A}_{1,ij}\bar{A}_{l,ik}\right)$$

- $\sum_{i\neq j\neq k}\bar{A}_{l;ij}\bar{A}_{l,ik} = 2\sum_{i<j<k}(\bar{A}_{l,ij}\bar{A}_{l,jk} + \bar{A}_{l,ik}\bar{A}_{l,kj} + \bar{A}_{l,ji}\bar{A}_{l,ki})$
- $\sum_{i\neq j\neq k}\bar{A}_{1,ij}\bar{A}_{l,ik} = \sum_{i<j<k}(\bar{A}_{1,ij}\bar{A}_{l,jk} + \bar{A}_{1,kj}\bar{A}_{l,ji} + \bar{A}_{1,ik}\bar{A}_{l,kj} + \bar{A}_{1,jk}\bar{A}_{l,ki} + \bar{A}_{1,ij}\bar{A}_{l,ik} + \bar{A}_{1,ik}\bar{A}_{l,ij})$

$$\sigma_n^2 = (1+o_P(1))\left(\frac{2(L-1)^2}{\rho_1^2\|W_1\|_1^2} + \sum_{l=2}^{L}\frac{2}{\rho_l^2\|W_l\|_1^2} + \sum_{l=2}^{L}\frac{4}{\rho_1\|W_1\|_1\rho_l\|W_l\|_1}\right)$$

Define $T_l, T_{1,l}$

$$T_{l;t} = \frac{\sqrt{2}}{\rho_l\|W\|_1}\sum_{i<j<k\leq t}(\bar{A}_{l,ij}\bar{A}_{l,jk} + \bar{A}_{l,ik}\bar{A}_{l,kj} + \bar{A}_{l,ji}\bar{A}_{l,ki})$$

$$T_{1,l;t} = \frac{1}{\sqrt{\rho_1\rho_l}\|W\|_1}\sum_{i<j<k\leq t}(\bar{A}_{1,ij}\bar{A}_{l,jk} + \bar{A}_{1,kj}\bar{A}_{l,ji} + \bar{A}_{1,ik}\bar{A}_{l,kj} + \bar{A}_{1,jk}\bar{A}_{l,ki} + \bar{A}_{1,ij}\bar{A}_{l,ki} + \bar{A}_{1,ik}\bar{A}_{l,ij})$$

Obtain Zn

$$Z_n = \frac{1}{\sigma_n}\left(\frac{\sqrt{2}(L-1)\rho_1\|W\|_1}{P_1}T_{1;n} + \sum_{l=2}^{L}\frac{\sqrt{2}\rho_l\|W\|_1}{P_l}T_{l;n} - \sum_{l=2}^{L}\frac{2\sqrt{\rho_1\rho_l}\|W\|_1^2}{\sqrt{P_1P_l}}T_{1,l;n}\right)$$

Asymptotic Distribution of D_n

Linear combination of random variables

$Y_n = \sum_{l=1}^{L} \lambda_l T_{l;n} + \sum_{l=2}^{L} \lambda_{1,l} T_{1,l;n}$

Specify a case L=3

$Y_t = \lambda_1 T_{1;t} + \lambda_2 T_{2;t} + \lambda_3 T_{3;t} + \lambda_4 T_{1,2;t} + \lambda_5 T_{1,3;t}$

for $3 \leq t \leq n$, and $Y_2 = 0$. Then $\{Y_t\}_{t=2}^{n}$ is a martingale.

$X_t = Y_t - Y_{t-1}$;

$X_t = \lambda_1 R_{1;t} + \lambda_2 R_{2;t} + \lambda_3 R_{3;t} + \lambda_4 R_{1,2;t} + \lambda_5 R_{1,3;t}$

where

$$R_{l;t} = \frac{\sqrt{2}}{\rho_l \|W\|_1} \sum_{i<j<t} (\bar{A}_{l,ij}\bar{A}_{l,jt} + \bar{A}_{l,it}\bar{A}_{l,tj} + \bar{A}_{l,ji}\bar{A}_{l,it}),$$

$$R_{1,l;t} = \frac{1}{\sqrt{\rho_1 \rho_l}\|W\|_1} \sum_{i<j<t} (\bar{A}_{1,ij}\bar{A}_{l,jt} + \bar{A}_{1,tj}\bar{A}_{l,ji} + \bar{A}_{1,it}\bar{A}_{l,tj} + \bar{A}_{1,jt}\bar{A}_{l,ti} + \bar{A}_{1,ji}\bar{A}_{l,it} + \bar{A}_{1,ti}\bar{A}_{l,ij}).$$

Then $\mathbb{E}[X_t | F_{t-1}] = 0$.

Hence, $\{X_t\}_{t=3}^{n}$ is a martingale difference.

Next is to prove the convergence of $\{X_t\}_{t=3}^{n}$

Asymptotic Distribution of D_n

The proof proceeds by verifying the two conditions in Proposition 4.2 prove assumptions (I) (II),

Proposition 4.2

Suppose that for every $n \in \mathbb{N}$ and $k_n \to \infty$ the random variables $X_{n,1}, \ldots, X_{n,k_n}$ are a martingale difference sequence relative to an arbitrary filtration $\mathcal{F}_{n,1} \subset \mathcal{F}_{n,2} \subset \ldots \subset \mathcal{F}_{n,k_n}$. If (I) $\sum_{i=1}^{k_n} \mathbb{E}(X_{n,i}^2 | \mathcal{F}_{n,i-1}) \to 1$ in probability, (II) $\sum_{i=1}^{k_n} \mathbb{E}(X_{n,i}^2 I[|X_{n,i}| > \epsilon] | \mathcal{F}_{n,i-1}) \to 0$ in probability for every $\epsilon > 0$, then $\sum_{i=1}^{k_n} X_{n,i} \to N(0,1)$ in distribution.

$\sum_{t=3}^{n} X_t \xrightarrow{D} N(0,1)$, and then $Y_n = \sum_{l=1}^{L} \lambda_l T_{l;n} + \sum_{l=2}^{L} \lambda_{1,l} T_{1,l;n} \xrightarrow{D} N(0,1)$.
By Cramer-Wold theorem, the vector of $[T_{1;n}, \ldots, T_{L;n}, T_{1,2;n}, \ldots, T_{1,L;n}]$ jointly converge to multivariate standard normal distribution with $2L-1$ dimensions.

Lemma 4.1

Under the assumption of Theorem 2, $T_{1;n}, \ldots, T_{L;n}$,$T_{1,2;n}$,$T_{1,3;n}$,$\ldots, T_{1,L;n}$ jointly converges in distribution to the $2L-1$ dimensional standard normal distribution.

By Lemma 4.1, the vector $(T_{1;n}, \ldots, T_{L;n}, T_{1,2;n}, T_{1,3;n}, \ldots, T_{1,L;n})$ jointly converges in distribution to the $2L-1$ dimensional standard normal distribution.

$$Z_n = \frac{1}{\sigma_n}\left(\frac{\sqrt{2}(L-1)\rho_1\|W\|_1}{P_1} T_{1;n} + \sum_{l=2}^{L} \frac{\sqrt{2}\rho_l\|W\|_1}{P_l} T_{l;n} - \sum_{l=2}^{L} \frac{2\sqrt{\rho_1\rho_l}\|W\|_1^2}{\sqrt{P_1 P_l}} T_{1,l;n}\right)$$

By the Slutsky's theorem, Z_n converges in distribution to the standard normal distribution. The proof is complete.

Example 1

- model assumptions
  $\|W_l\|_1 = \omega(1), \quad \|W_l\|_2 = 1, \quad \|W_l\|_4 = o(1), \quad \sum_i W_{1;i} W_{l;i} \leq 1 - \epsilon,$
  $\min_{1 \leq l \leq L}\{\rho_l\} = \omega(1)$, $\rho_l = o(\|W_l\|_1)$.

Example 1 (Yuan and Yao 2025)

Let $W_{l,i} = \frac{\lambda_l \sqrt{r}}{\sqrt{n}}$ for $1 \leq i \leq \frac{n}{r}$ and $W_{l,i} = \frac{\sqrt{\frac{r}{r-1}(1-\lambda_l^2)}}{\sqrt{n}}$ for $\frac{n}{r} < i \leq n$, $0 < \lambda_l \leq 1$ and $r > 1$.

$$\text{Then } \|W_l\|_1 = \Theta(\sqrt{n}), \quad \|W_l\|_2 = 1, \quad \|W_l\|_4^4 = O\left(\frac{1}{n}\right).$$

$$\sum_{i=1}^{n} W_{1,i} W_{l,i} = \lambda_1 \lambda_l + \sqrt{(1-\lambda_1^2)(1-\lambda_l^2)} \leq 1 - \epsilon, \text{ when } \lambda_1 = \lambda_l \text{ or } \lambda_1 \neq \lambda_l$$

If $\min_{1 \leq l \leq L}\{\rho_l\} = \omega(1)$, $\rho_l = o(\sqrt{n})$ such as $n^\tau, \tau < 0.5$, Theorem 1 and Theorem 2 holds.

Example 1

$\rho_l = n^{\tau_l}$, $\tau \in \{0.3, 0.2, 0.4\}$, $n \in \{200, 250, 300\}$
$\min_{1\le l\le L}\{\rho_l\} = \omega(1)$, $\rho_l = o(\sqrt{n})$

$\|W_l\|_1 = \sum w_{l,i} = \Theta(\sqrt{n})$,
$\|W_l\|_1 = \sum w_{l,i} = \frac{n}{r}.\frac{\lambda_l\sqrt{r}}{\sqrt{n}} + (n - \frac{n}{r}).\frac{\sqrt{\frac{r}{r-1}(1-\lambda_l^2)}}{\sqrt{n}} = \sqrt{n}\left[\lambda_l r^{-\frac{1}{2}} + (\frac{r}{1-r})^{-\frac{1}{2}}(1-\lambda_l^2)^{\frac{1}{2}}\right] = \sqrt{n}C = \Theta(\sqrt{n})$
$\|W_l\|_2 == (\sum w_{l,i}^2)^{\frac{1}{2}} = 1$,
$\|W_l\|_4^4 = (\sum w_{l,i}^4)^{\frac{1}{4}} = O\left(\frac{1}{n}\right)$
$\|W_l\|_4^4 = (\sum w_{l,i}^4)^{\frac{1}{4}} = \frac{n}{r}(\frac{\lambda_l\sqrt{n}}{\sqrt{n}})^4 + (n - \frac{n}{r})(\frac{\sqrt{\frac{r}{r-1}(1-\lambda_l^2)}}{\sqrt{n}})^4 = \frac{1}{n}[r\lambda_l^4 + (\frac{r}{r-1})(1-\lambda_l^2)^2]$
$\sum w_{1,i}w_{l,i} = \frac{n}{r}.\frac{\lambda_1\sqrt{r}}{\sqrt{n}}.\frac{\lambda_l\sqrt{r}}{\sqrt{n}} + (n - \frac{n}{r}).\frac{\sqrt{\frac{r}{r-1}(1-\lambda_1^2)}}{\sqrt{n}}.\frac{\sqrt{\frac{r}{r-1}(1-\lambda_l^2)}}{\sqrt{n}} = \lambda_1\lambda_l + \sqrt{(1-\lambda_1^2)(1-\lambda_l^2)}$
If $\lambda_1 = \lambda_l$, $\sum w_{1,i}w_{l,i} = 1$;
If $\lambda_1 \neq \lambda_l$, $\sum w_{1,i}w_{l,i} \le 1 - \epsilon$;

Example 2

- model assumptions
  $\|W_l\|_1 = \omega(1), \quad \|W_l\|_2 = 1, \quad \|W_l\|_4 = o(1), \quad \sum_i W_{1;i} W_{l;i} \leq 1 - \epsilon,$
  $\min_{1 \leq l \leq L}\{\rho_l\} = \omega(1),\ \rho_l = o(\|W_l\|_1).$

Example 2 (Yuan and Yao 2025)

Let $W_{l,i} = \frac{i^{\beta_l}}{\sqrt{S_{n,2\beta_l}}}, \quad S_{n,2\beta_l} = \sum_{i=1}^n i^{2\beta_l},\ \beta_l > 0.$

$$\text{Then } \|W_l\|_1 = \frac{1}{\sqrt{S_{n,2\beta_l}}} \sum_{i=1}^n i^{\beta_l} = \Theta\left(\sqrt{n}\right), \quad \|W_l\|_2^2 = 1, \quad \|W_l\|_4^4 = O\left(\frac{1}{n}\right).$$

$$\sum_{i=1}^n W_{1,i} W_{l,i} = \sum_{i=1}^n \frac{i^{\beta_1}}{\sqrt{S_{n,2\beta_1}}} \frac{i^{\beta_l}}{\sqrt{S_{n,2\beta_l}}} = (1 + o(1)) \frac{\sqrt{(2\beta_1+1)(2\beta_l+1)}}{\beta_1+\beta_l+1} \leq 1 - \epsilon,$$
$$\text{when } \beta_l = \beta_1 \text{ or } \beta_l \neq \beta_1.$$

If $\min_{1 \leq l \leq L}\{\rho_l\} = \omega(1),\ \rho_l = o(\sqrt{n})$ such as $n^\tau, \tau < 0.5$, Theorem 1 and Theorem 2 holds.

Example 2

$\rho_l = n^{\tau_l}$, $\tau \in \{0.3, 0.2, 0.4\}$, $n \in \{200, 250, 300\}$
$\min_{1 \le l \le L}\{\rho_l\} = \omega(1)$, $\rho_l = o(\sqrt{n})$

$\|W_l\|_1 = \sum w_{l,i} = \Theta(\sqrt{n})$,
$\|W_l\|_2 == (\sum w_{l,i}^2)^{\frac{1}{2}} = 1$,
$\|W_l\|_4^4 = (\sum w_{l,i}^4)^{\frac{1}{4}} = O\left(\frac{1}{n}\right)$

$$\sum w_{1,i} w_{l,i} = \frac{1^{\beta_1} 1^{\beta_l}}{\sqrt{\sum j^{2\beta_1}} . \sqrt{\sum j^{2\beta_l}}} + \frac{2^{\beta_1} 2^{\beta_l}}{\sqrt{\sum j^{2\beta_1}} . \sqrt{\sum j^{2\beta_l}}} + \cdots + \frac{n^{\beta_1} n^{\beta_l}}{\sqrt{\sum j^{2\beta_1}} . \sqrt{\sum j^{2\beta_l}}} =$$
$$\frac{1}{\sqrt{\sum j^{2\beta_1}} . \sqrt{\sum j^{2\beta_l}}}(1^{\beta_1+\beta_l} + 2^{\beta_1+\beta_l} + \cdots + n^{\beta_1+\beta_l}) = \frac{1}{\sqrt{\frac{n^{2\beta_1+1}}{2\beta_1+1}} . \sqrt{\frac{n^{2\beta_1+1}}{2\beta_l+1}}} . \frac{n^{\beta_1+\beta_l+1}}{\beta_1+\beta_l+1} = \frac{\sqrt{(2\beta_1+1)(2\beta_l+1)}}{\beta_1+\beta_l+1}$$

If $\lambda_1 = \lambda_l$, $\sum w_{1,i} w_{l,i} = 1$;
If $\lambda_1 \neq \lambda_l$, $\sum w_{1,i} w_{l,i} \le 1 - \epsilon$;

Subspace, Dimension, Rank

- A vector space $\mathbb{R}^n$ is a set V of vectors $(x_1, x_2, \ldots, x_n)$ along with an addition on V and a scalar multiplication such that properties such as commutativity, associativity, distributive, etc. hold.
- **Subspace**: a subset U of V is called a subspace of V if U is also a vector space.
- A **basis** of V is a list of vectors in V that is **linearly independent** and **spans** V. There are n vectors in the standard basis of $\mathbb{R}^n$.
- The set of all linear combinations of a list of vectors $v_1, v_2, \ldots, v_m$ in V is called the span of $v_1, v_2, \ldots, v_m$, denoted as span$\{v_1, v_2, \ldots, v_m\}$. The span of a list of vectors in V is the smallest subspace of V containing all the vectors in the list.
- length of linearly independent list $\leq$ length of spanning list. (if a list of vector in span are linearly independent, then equal.)
- The **dimension** of a finite-dimensional vector space is the length of any basis (linearly independent and spans the space) of the vector space.
  The dimension of subspace form by span of $v_1, v_2, \ldots, v_m$ is the number of the vectors in the list.
- Suppose matrix A is m-by-n matrix with entries in F, the **row rank** of A is the dimension of the span of the linearly independent rows of A in $\mathbb{R}^n$; the **column rank** of A is the dimension of the span of the linearly independent columns of A in $\mathbb{R}^m$. The rank of a matrix $A \in \mathbb{R}^{m,n}$ is the column rank of A. the linearly independent columns form a basis for the column space.

Subspace, Dimension, Rank

- **Linear map**: Suppose $v_1, v_2, \ldots, v_n$ is a basis of V and $w_1, w_2, \ldots, w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that $Tv_j = w_j$ for each $j = 1, \ldots, n$. The set of all linear maps from V to W is denoted as $\mathcal{L}(V, W)$. If a linear map from a vector space to itself, $\mathcal{L}(V)$.
  $\mathcal{L}(V, W)$ is a vector space and can be represented as a matrix $\mathcal{M}(T)$.
  Map $(x, y) \in \mathbb{R}^2$ to $(x + 3y, 2x + 5y, 7x + 9y) \in \mathbb{R}^3$ by $\mathcal{M}(T) = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{bmatrix}$
- Suppose $T \in \mathcal{L}(V)$. A **subspace** U of V is called **invariant** under T if $u \in U$ implies $Tu \in U$.
  Take any $v \in V$ with $v \neq 0$ and let U equal the set of all scalar multiples of v: $U = \{\lambda v : \lambda \in \mathbb{R}\} = span(v)$.
  Then U is a 1-dimensional subspace of V. If U is invariant under an operator $T \in \mathcal{L}(V)$, then $Tv \in U$, and hence there is a scalar $\lambda \in \mathbb{R}$ such that $Tv = \lambda v$. Conversely, if $Tv = \lambda v$ for some $\lambda \in \mathbb{R}$, then $span\{v\}$ is a 1-dimensional subspace of V invariant under T.
- Let $T \in \mathcal{L}(V)$. Suppose $\lambda_1, \lambda_2, \ldots, \lambda_m$ are distinct eigenvalues of T and $v_1, v_2, \ldots, v_m$ are corresponding eigenvectors. Then $v_1, v_2, \ldots, v_m$ is linearly independent. $span\{v_1, v_2, \ldots, v_m\}$ form a subspace of T with d-dimensional.
  $Tx = \lambda x$ to $TU = U\Lambda$, $T = U\Lambda U^T$, columns $v_1, v_2, \ldots, v_m$ in U are orthonormal eignevectors and form an orthonormal basis of $\mathbb{R}^m$. $span\{v_1, v_2, \ldots, v_d\}$, $d \leq m$ is a d-dimensional subspace in $\mathbb{R}^n$ invariant under T.

Empirical Likelihood

Definition 4 Empirical Cumulative Distribution Function (ECDF)

Let $X_1, X_2, ..., X_n \in \mathbb{R}$. The empirical cumulative distribution function (ECDF) of $X_1, X_2, ..., X_n$ is

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{X_i \leq x}, \text{ for } -\infty < x < \infty$$

Definition 5 Nonparametric Likelihood

Given $X_1, X_2, ..., X_n \in \mathbb{R}$, assumed independent with common CDF F_0, the nonparametric likelihood of the CDF F is

$$L(F) = \prod_{i=1}^{n} [F(X_i) - F(X_i-)]$$

Theorem 3 ECDF is the Nonparametric MLE of F

Let $X_1, X_2, ..., X_n \in \mathbb{R}$ be independent random variables with a common CDF F_0. Let F_n be their ECDF and let F be any CDF. If $F \neq F_n$, then $L(F) < L(F_n)$.

Empirical Likelihood

Definition 6 Empirical Likelihood Ratio function

$R(F) = \frac{L(F)}{L(F_n)} = \prod_{i=1}^{n} nw_i$, where $w_i \geq 0, \sum_{i=1}^{n} w_i \leq 1$.

Definition 7 Empirical Likelihood Ratio function for μ

$R(\mu) = max\{\prod_{i=1}^{n} nw_i | \sum_{i=1}^{n} w_i X_i = \mu, w_i \geq 0, \sum_{i=1}^{n} w_i = 1\}$

Theorem 4 Empirical Likelihood Theorem

Let $X_1, X_2, ..., X_n \in \mathbb{R}^d$ be independent random variables with a common CDF F_0. $\mu_0 = E(X_i)$, finite variance covariance matrix V_0 of rank $q > 0$. Then $-2log(\mathcal{R}(\mu_0))$ converges in distribution to $\chi^2_{df=q}$ as $n \to \infty$.

$$R(\mu) = \frac{MLEH_0}{MLEH_1}$$
$$\text{under } H_0 : \mu = \sum_{i=1}^{n} w_i X_i = 0 \quad \text{under } H_1 : \mu = \sum_{i=1}^{n} \frac{1}{n} X_i.$$
$$\text{small } R, \text{ against } H_0. \text{ large } -2logR, \text{ against } H_0.$$

Weighted Degree Difference Data:

$$X_i = \sum_{l=2}^{L} \left[\left(\frac{d_{1,i}}{\sqrt{P_1}} - \frac{d_{l,i}}{\sqrt{P_l}} \right)^2 - \frac{d_1}{P_1} - \frac{d_l}{P_l} \right], 1 \leq i \leq n, \text{ Where } E[X_i] = \mu, 0 < Var[X_i] < \infty$$

EL test

hypothesis

By Approach 1 we know that $D_n \xrightarrow{D} N(0,1)$ as $n \to \infty$. An equivalent hypothesis testing by Empirical Likelihood Ratio is to text whether $E[X_i] = 0$

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \neq \mu_0, \text{ where } \mu_0 = 0.$$

Under H_0, the graphs $A_1, A_2, \ldots, A_L$ have the same degree-correction parameters. Under H_1, there exist at least two graphs such that their degree-correction parameters are different.

test statistics

Empirical Likelihood Ratio function for μ_0

$$R_n = max\{\prod_{i=1}^{n} nw_i | \sum_{i=1}^{n} w_i X_i = 0, w_i \geq 0, \sum_{i=1}^{n} w_i = 1\}$$

Theorem 5 Convergence of $-2\log R_n$

Let $\mu_0 = E(X_i)$ and suppose that $0 < Var(X_i) < \infty$. $-2\log R_n$ converges in distribution to $\chi^2_{df=1}$ as $n \to \infty$.

$$\alpha = P(-2\log R_n > C_n \| \mu_0 = 0)$$
$$\beta = P(-2\log R_n > C_n \| \mu_0 \neq 0) \xrightarrow{P} 1$$

The EL test is defined as:

$$\text{Reject } H_0 \text{ if } -2\log R_n > \chi^2_{df=1,1-\alpha}$$